

Voiceless Arabic vowels recognition using facial EMG

Luay Fraiwan · Khaldon Lweesy · Ayat Al-Nemrawi ·
Sondos Addabass · Rasha Saifan

Received: 3 August 2010 / Accepted: 15 February 2011 / Published online: 16 March 2011
© International Federation for Medical and Biological Engineering 2011

Abstract This work attempts to recognize the Arabic vowels based on facial electromyograph (EMG) signals, to be used for people with speech impairment and for human computer interface. Vowels were selected since they are the most difficult letters to recognize by people in Arabic language. Twenty subjects (7 females and 13 males) were asked to pronounce three Arabic vowels continuously in a random order. Facial EMG signals were recorded over three channels from the three main facial muscles that are responsible for speech. The EMG signals are then pre-processed to eliminate noise and interference signals. Segmentation procedure was implemented to extract the time event that corresponds to each vowel based on a moving standard deviation window. The accuracy of the segmentation procedure was found to be 94%. The recognition of the vowels was carried out by extracting features from the EMG in three domains: the temporal, the spectral, and the time frequency using the wavelet packet transform. Classification of the extracted features was then finally performed using different classification methods implemented in the WEKA software. The random forest classifier with time frequency features showed the best performance with an accuracy of 77% evaluated using a 10-fold cross-validation.

Keywords Facial EMG · Wavelet packets · Feature selection · Random forest classifier

1 Introduction

Voiceless speech recognition has recently found many applications such as human computer interface and helping people with speech impairment. Human computer interface has been applied and used in many systems where some of these systems employ the speech recognition for human computer interaction. However, they are not suitable for people with speech impairment and they do not work efficiently in a noisy environment or when there are people in the vicinity of the operator [5, 14, 18, 21, 32]. A voiceless speech system based on surface electromyograph (EMG) signals from the facial muscles can be useful to overcome these drawbacks. Such a system can be defined as the decoding of language speech non-vocally based solely on the EMG signals from the facial muscles [6]. It consists mainly of the following major parts: data acquisition, data processing, feature extraction (data parameterization), and a classification procedure. The data acquisition system depends on the recognition of the shape of the face and the related muscle activity. This is done by detecting the facial movement, which can be achieved by visual sensing, mechanical sensing, or electrical sensing. Visual sensing uses a camera to record the shape of the face [1, 3, 20, 24, 26], while mechanical sensing uses mechanical sensors to detect the movement of the face during speech, while electrical sensing records the facial muscle activity using the EMG signal [2]. Regardless of the sensing method, the recorded data need to be processed to eliminate noise and interference and to identify the time events. This step is important as it largely affects the accuracy of the speech recognition system. The next part is feature extraction from the processed data which has the form of a time series. It is not efficient to use the processed data for classifications, instead parameters or features are extracted to reduce the

L. Fraiwan (✉) · K. Lweesy · A. Al-Nemrawi ·
S. Addabass · R. Saifan
Biomedical Engineering Department, Jordan University
of Science & Technology, PO Box 3030,
Irbid 22110, Jordan
e-mail: fraiwan@just.edu.jo

size of the data and have a better representation. The feature extraction can be done in three domains: temporal, spectral, and time–frequency. The extraction of features in the temporal domain identifies the parameters that characterize the temporal structure of the recorded signal relative to the specific muscular movement. This is done by using information in the time domain as feature values without doing any further transformation [25, 35]. In the spectral domain, attributes of the recorded data are extracted from the spectrum of the EMG signals using Fourier transform-based methods [31]. In the third technique, recorded data are transformed into a time–frequency representation using methods such as short time Fourier transform (STFT) and wavelet transform, with information in time–frequency domain used as features [9, 12, 34]. Classification is the last step where numerous methods have found success in similar applications such as artificial neural networks, fuzzy logic, linear discriminant analysis, and classification trees [7, 15, 19].

Although Arabic language is a widely used, it has yet to be extensively studied in the area of speech and voiceless recognition systems. The purpose of this study is to build a reliable and accurate system for voiceless speech recognition for Arabic language vowels as they are the most difficult letters to recognize. Different methods for feature extraction in temporal, spectral, and time–frequency domains were implemented and explored. Moreover, many classifiers were used and tested to recognize the Arabic vowels.

2 Methods

The entire voiceless Arabic vowels recognition system is shown in Fig. 1. The system consists of six major parts: facial EMG data acquisition, pre-processing of the recorded EMG data, segmentation to extract the time event of each vowel, feature extraction, feature selection, and classification.

2.1 Facial EMG data acquisition

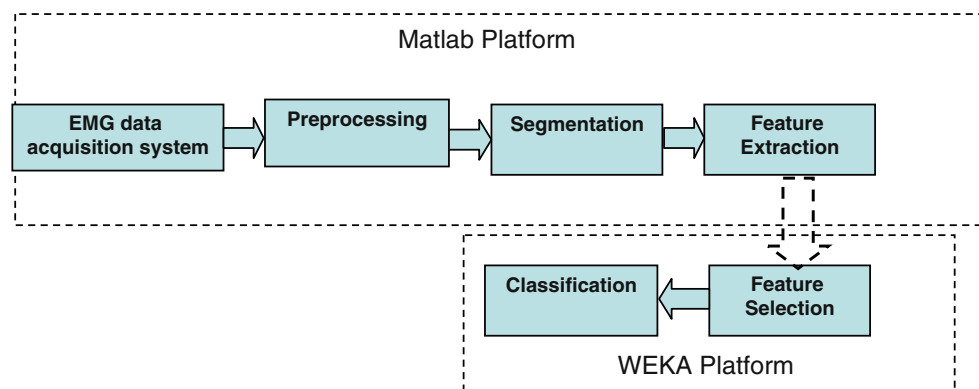
Facial EMG data from 20 subjects, 7 females and 13 males, were recorded in this study. They were asked to say the Arabic vowels ‘ا’, ‘ي’, and ‘و’ non-acoustically 15 times in a random order. A three channel facial EMG system was used to collect data from the subjects. Three pairs of pre-gelled Ag/AgCl electrodes were used to record the surface EMG signal from three facial muscles that are mainly responsible for speech [13]: Orbicularis oris (channel 1) which is the sphincter muscle around the mouth, Triangularis (channel 2) which originates in the mandible and platysma, and inserts in both the skin and the orbicular muscle at the corner of the mouth, and Risorius (channel 3) which originates in the fascia of the masseter below the zygomatic arch and inserts in the skin near the corner of the mouth. The positions of the electrodes were selected to have a good and stable EMG signal. They were placed longitudinally with respect to the long axis of the muscle as shown in Fig. 2. The reference electrode was placed on the left hand wrist [16]. Data from the three channels were amplified and recorded using a signal conditioner and a data acquisition system from National Instruments-USA with a sampling rate of 4050 Hz.

2.2 Pre-processing

The purpose of the pre-processing step, illustrated in Fig. 3, was to prepare the recorded EMG signal for the segmentation and feature the extraction steps. This part includes: eliminating noise contaminating the signal, removing other signals and artifacts that interfere with the EMG signal and smoothing the EMG data. The major source of noise was found to be the 50 Hz power line interference, and an infinite impulse response (IIR) notch filter was used to eliminate this noise.

A normal EMG signal has a frequency response in the range of 2–500 Hz [8, 30]. In this frequency range, signals such as ocular electrooculograph (EOG) and electrocardiograph (ECG) interfere with the EMG signal. Moreover,

Fig. 1 Voiceless Arabic vowel recognition system



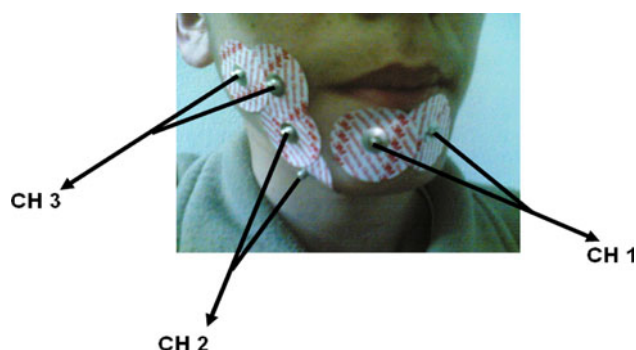


Fig. 2 Placement of the facial EMG electrodes

artifacts like air flow, head movement, and cables movement appear in this frequency range. In this work, each of the above-mentioned interfering signals and artifacts were evaluated separately and were found to be attenuated largely at frequencies higher than 20 Hz. Therefore, a Butterworth band pass IIR filter with a bandwidth of 20–500 Hz was used to eliminate the interfering signals and the artifacts. The Butterworth filter was used since it has smooth pass and stop bands.

Smoothing the data was done using a Savitzky–Golay filter [22], a highly effective filter in removing Gaussian noise and impulse noise. The Savitzky–Golay filter essentially performs a local polynomial regression of degree k on a distribution of at least $k + 1$ equally spaced points to determine the smoothed value for each point. The main advantage of this approach is that it preserves features of the distribution such as relative maxima, relative minima, and width, which are usually flattened by other averaging techniques. The final stage of the smoothing procedure was cubing of the EMG signal (raising the EMG signal to the power 3) to suppress low signal values. The cubed EMG signal was only deployed in the segmentation procedure.

More processing was performed to remove hardware oscillations, which appear when the device is switched on, until it reaches its stable state. The hardware oscillation was removed by discarding the first 200 ms of the recorded

signal. Then the signal was normalized by dividing the EMG signal by the maximum value, and the DC bias was removed by subtracting the mean from the EMG signal.

2.3 EMG segmentation

Segmentation was used to identify the start and the end times of the spoken vowel from a silence background in a sequence of spoken vowels [28, 29]. It is critical to the success of the eventual classifier that the correct part of the EMG signal is extracted for analysis. Incorrectly designating the start and the end times of speaking results in the inclusion of noise, which affects directly the accuracy of the classifier in the next steps. The segmentation algorithm followed in this study is shown in Fig. 4, where the cubed EMG signal was deployed [23, 27]. It was based on the exchange of the cubed EMG signal with another signal that is correlated to it and has clear events that can be easily detected. This signal was chosen to be the standard deviation (std) of a moving window with a specified duration. Then a binary signal was generated by applying a threshold to the std signal. This binary signal corresponds to whether an event exists or not. The threshold was chosen as follows:

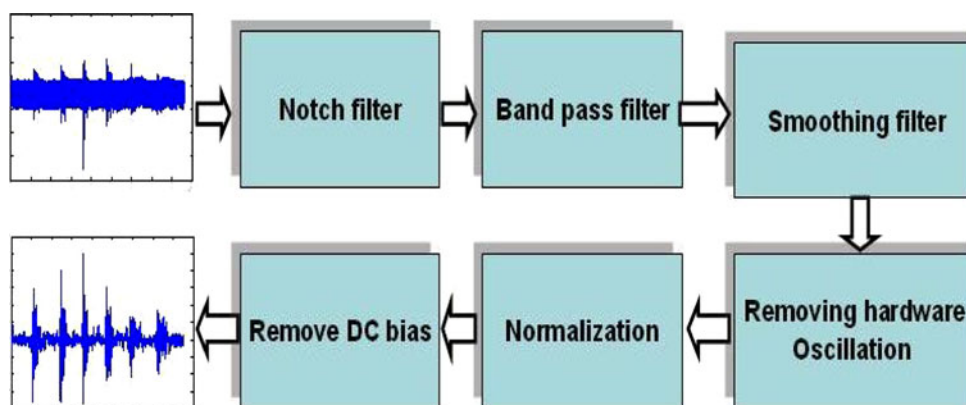
$$Th = \text{mean}(\text{rest}) + \mu \text{std}(\text{rest}), \quad (1)$$

where Th is the threshold value, rest is the first second of recording before any time event, and μ is the sensitivity which equals 6, the value that gave the best performance of the segmentation algorithm based on experiments performed on the collected data. The maximum length of a time event was set to be 2 s. This was based on separate recordings of the vowels where it was found that in all cases the average time of each vowel did not exceed 2 s.

2.4 Feature extraction

After identifying each time event separately, the corresponding EMG signal was obtained, and features were

Fig. 3 Preprocessing procedure



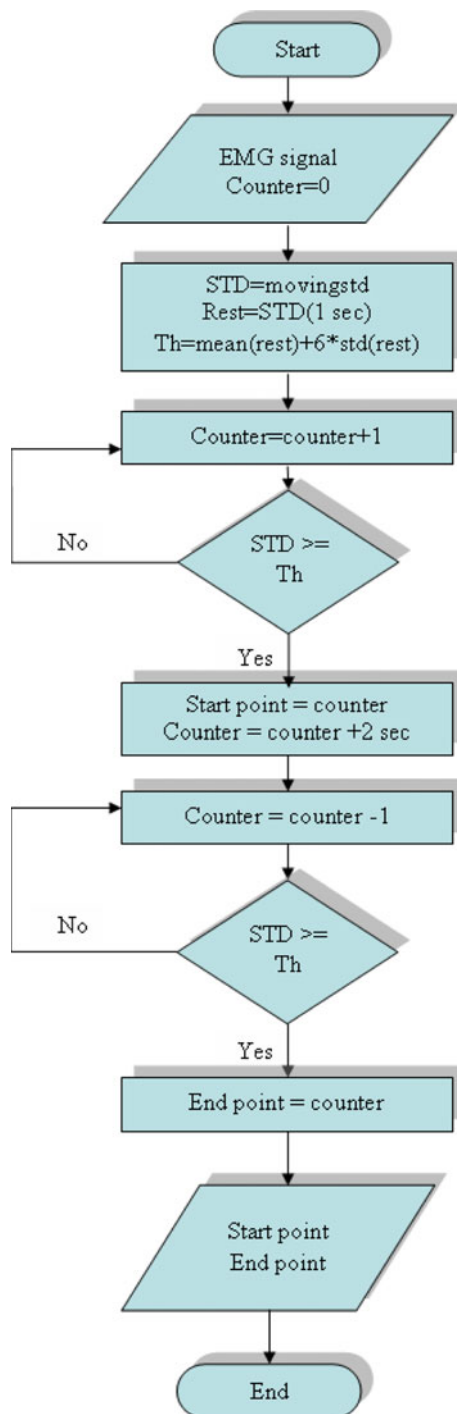


Fig. 4 Segmentation algorithm

extracted from the EMG signal. The feature extraction step has the greatest impact on the ability of a classifier to successfully associate the raw data with its class. In this study, features were extracted in three different domains: temporal, spectral, and time–frequency using wavelet packets.

2.4.1 Temporal features

The temporal approach identifies the attributes of the raw EMG signal that characterize its temporal structure relative to a specific muscular function. The temporal features used in this work were [25, 35]:

- Mean absolute value (MAV), which is calculated by taking the average of the absolute value of the EMG signal during the specified event for the three recorded channels separately.
- Zero crossings (ZC), the number of times the EMG signal crosses the zero line.
- The variance (VAR) of the EMG signal.
- Slope sign changes (SSC), which tracks the number of changes between positive and negative slopes among three subsequent EMG samples.
- Waveform duration (Td); the duration of the time events, in this work, it was the number of samples in the time events.
- The energy (E) of the EMG signal in each channel.

2.4.2 Spectral features

The spectral approach identifies the attributes of the raw EMG signal from its spectrum in the frequency domain [31]. The extracted features were:

- Frequency ratio which employs the fast Fourier transform (FFT) to determine the ratio of the magnitude of the lowest frequency to the magnitude of the highest frequency for the three channels.
- Spectral moments which were calculated based on the power spectral density (PSD). The PSD, $P(f)$, was calculated using the FFT of the EMG signal $s(t)$ of length N as follows:

$$P(f) = \frac{1}{N} |S(f)|^2, \quad (2)$$

where $S(f)$ is the Fourier transform of the EMG signal $s(t)$ and N is the length of the EMG signal (the duration of the EMG signal during the vowel speech). The spectral moment M of order m is defined as:

$$M_m = \int P(f) f^m df, \quad (3)$$

The spectral moments were calculated up to the fifth order ($m = 0, 1, \dots, 5$)

- Spectral magnitude average, determined by averaging the PSD at fixed frequency intervals:

$$\hat{P}_n = \frac{1}{\Delta} \int_{f_{n-1}}^{f_n} P(f) df \quad (4)$$

where $\hat{P}_n$ is the spectral magnitude average at the frequency interval n and Δ is the fixed frequency interval. In this study, the signal was considered as one interval, $n = 1$.

2.4.3 Time–frequency features and wavelet packets

The wavelet packet transform (WPT) is a general expansion of the discrete wavelet transform (DWT). WPT has been successfully used in different fields as a feature extraction method [4, 10, 17]. It possesses the ability to locate features both in time and frequency domains. With WPT, the signal is decomposed in an approximation space and details space. The approximation space is done by a scaling function and the details space is done by a mother wavelet function. Each of the two spaces is decomposed again into approximation subspace and details subspace. The features extracted from each subspace in the frequency range of interest for each EMG channel were:

- the standard deviation (std),
- the frequency ratio defined as the ratio of the magnitude of the highest wavelet coefficient to the magnitude of the lowest wavelet coefficient,
- the energy,
- the power of the subspace normalized to the total power of the signal.

2.5 Classification methods

There are many classification algorithms that can be used for EMG pattern recognition such as neural networks, fuzzy logic, regression trees, and linear discriminant analysis [7, 15, 19]. These methods have a varying degree of success depending on the type of application. In this work, WEKA (the University of Waikato, New Zealand), an open source and machine learning software, was used for the classification and the evaluation of the performance of many algorithms for the different extracted features as mentioned earlier [17]. Before doing the classification step, feature selection was performed. Some of the features used by the classifier may be redundant and may be a source of error in the classification procedure. Therefore, a feature selection technique was implemented using the gain ratio method, which uses the information gain of each feature in the selection process [11, 33]. If f is the set of all features and Ex is the set of all training data, the information gain (IG) for a feature $a \in f$ is defined as:

$$IG = H(Ex) - \sum_{v \in \text{values}(a)} \frac{|\{x \in Ex | \text{value}(x, a) = v\}|}{|Ex|} H(|\{x \in Ex | \text{value}(x, a) = v\}|), \quad (5)$$

where $\text{value}(x, a)$ defines the value of a specific x for feature a , and H specifies the entropy, which for a random variable X with possible values $\{x_1, x_2, \dots, x_n\}$ is given by:

$$H(X) = - \sum_{i=1}^n p(x_i) \log p(x_i). \quad (6)$$

where p denotes the probability mass function of X . The gain ratio (GR) is defined as:

$$GR(X) = \frac{IG}{-\sum_{i=1}^n \log p(x_i)}. \quad (7)$$

Only features with a gain ratio greater than zero was selected. The number of time–frequency features (wavelet packets) selected was 14 out 48, temporal features 13 out 18, and spectral features 7 out of 9.

3 Results

The EMG data from the three channel system were recorded and imported into Matlab using the code Softscope. In the pre-processing stage, an IIR notch filter with a quality factor of 12 was used to remove the power line interference, and a Butterworth band pass filter (20–500 Hz) was used to remove the interferences and artifacts from other sources. The smoothing of the signals was done using first-order Savitzky–Golay filter with a smoothing window of 100 ms. For illustration purposes, Fig. 5 shows the filtration of the EOG signal.

The segmentation procedure was done for one channel and generalized to the other two channels since the onset and the offset times are the same. In this work, channel one (EMG from the Orbicularis oris muscle) was chosen. The standard deviation moving window signal was calculated based on window duration of 0.25 s. Figure 6 shows the EMG signal and the corresponding standard deviation moving window signal. The results of the segmentation procedure algorithm mentioned earlier are shown in Fig. 7. For the 300 vowels recorded from the subjects (20 subjects with 15 instances), the accuracy of the segmentation procedure tested against the manual annotation made during the recording procedure was 94%.

The EMG segment was obtained from the above procedure, and features were extracted using the three techniques in the temporal, spectral, and time–frequency domains. A features vector for each technique was formed. The length of the temporal features vector was 18 (6 from

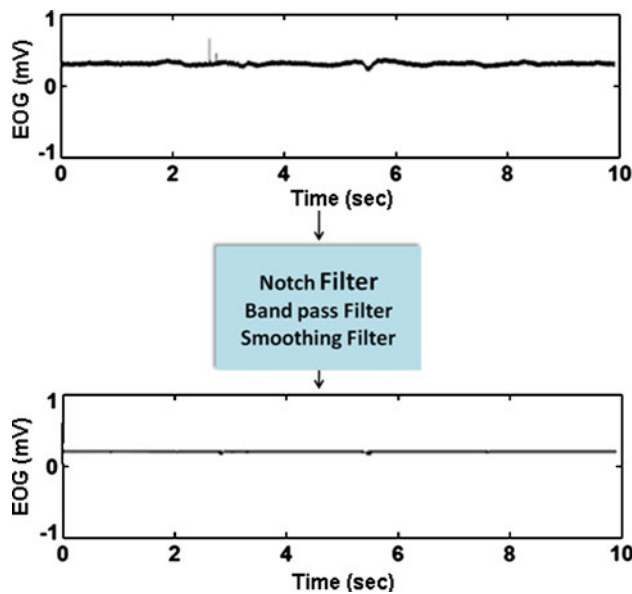


Fig. 5 Filtration of the EOG interference

each channel). The spectral features vector was formed from frequency ratio, spectral magnitude average (the average was calculated as one frequency interval), and the first five spectral moments. The total length of the spectral moving average was 21.

The time–frequency features were calculated based on the wavelet packets decomposed up to four levels using the Daubechies mother wavelet of order 20 (db20). The features values were calculated of each subspace in the frequency range of interest as shown in Fig. 8. The number of sub-bands was 7 which made the length of the features vector 28 for each channel, 84 for the three channels.

The features for the data recorded from the subjects were calculated and used to form a data set of 300 instances (100 for each vowel). The features data were fed into WEKA classification software. Testing was done based on

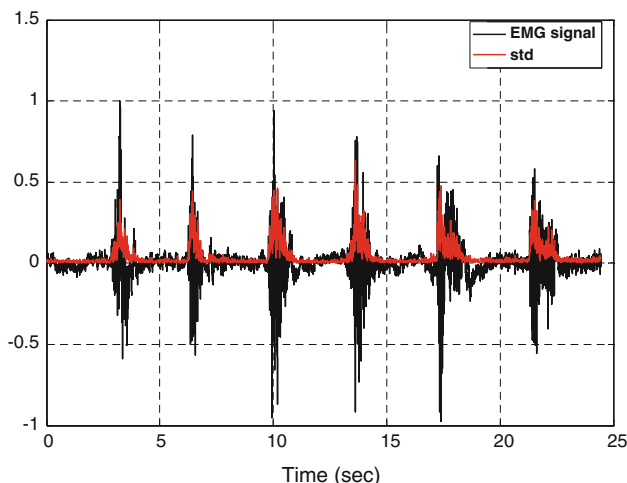


Fig. 6 EMG signal and the standard deviation moving window signal

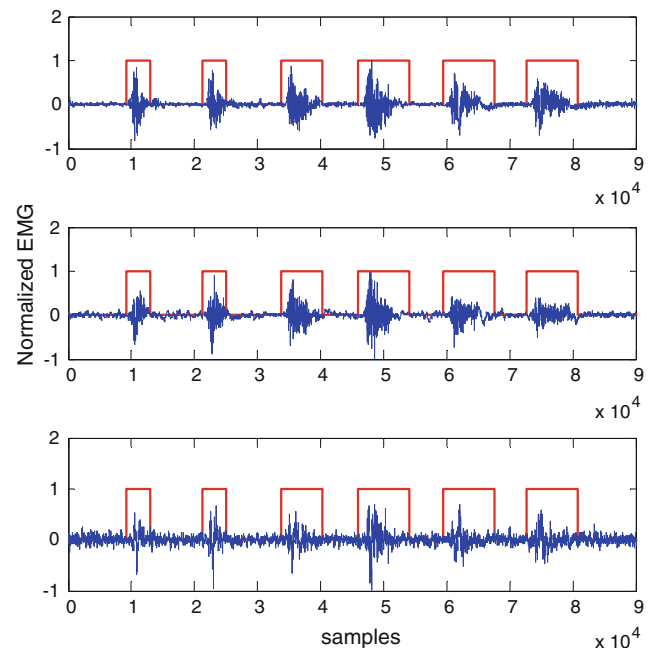


Fig. 7 Segmentation results

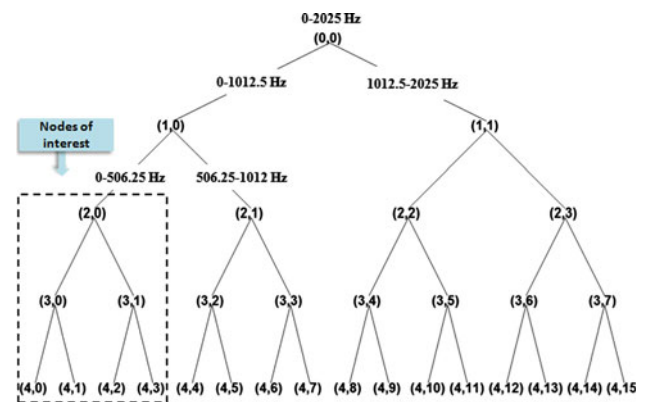


Fig. 8 Four-level Wavelet packets decomposition tree

a 10-fold cross validation. Table 1 shows the results of the classification procedure using a variety of classifiers for the three methods of feature extraction. The highest accuracy was obtained using the random forest classifier for the three feature extraction methods, with wavelets packets features having the highest accuracy. Table 2 shows the confusion matrix for the random forest classifier with wavelets packet features.

Analysis of the wavelet packet features used in the classification procedure was done as shown in Table 3. The table shows the wavelet packets features along with their gain ratio values used for the classification procedure using the random forest classifier (the remaining features have zero gain ratio value). The most used node was node (0,4) from the recorded three channels with a frequency band of (0–127 Hz).

Table 1 Accuracy of the classification procedure

Classifier	Temporal features (%)	Spectral features (%)	WP features (%)
Ridor	60.3	58	59.3
J48	63	58	65.3
FT	60.3	60	72
NNge	65.3	68.3	58.3
FLR	64.7	66.3	43.3
Decision table	56	50	62
Random forest	71	70.3	77
Random tree	58.7	64.7	50.7
REP tree	57.7	58	61.7
Average accuracy	61.89	61.51	61.07

Table 2 Confusion matrix for random forest classifier and WP features

Vowel	Classified as:			Accuracy
	ا	ي	و	
ا	77	17	6	77%
ي	21	72	7	72%
و	9	9	82	82%

4 Discussion

A complete system for Arabic vowels recognition based on three EMG channels was implemented. The implementation was done in five steps: facial EMG data acquisition, pre-processing, segmentation, feature extraction, and classification. The pre-processing stage was efficient in removing noise, artifacts, and interferences, as well as smoothing the EMG signal. Normalization was performed

Table 3 Gain ratio for the different features used in the classification procedure

Feature	Node	Channel	Gain ratio
std	(4,0)	3	0.2432
F ratio	(4,0)	1	0.2260
Power	(4,0)	1	0.1958
Energy	(4,0)	2	0.1546
Power	(4,0)	2	0.1339
std	(4,0)	2	0.1191
Power	(4,2)	3	0.1167
std	(4,0)	1	0.1068
std	(4,1)	3	0.0923
Energy	(4,3)	2	0.0921
std	(4,3)	3	0.0839
Power	(4,3)	3	0.0793
Energy	(4,1)	3	0.0766
Energy	(4,0)	3	0.0736

so that the classification procedure is independent of the subject and the EMG data acquisition system.

A threshold based segmentation method was applied to extract the time events from an alternative standard deviation moving window signal that is correlated to the original EMG signal. The challenge was to select the appropriate threshold value from the rest interval. This method has shown to have a good performance in extracting time event from a sequence of events where the accuracy was found to be 94%. Many other methods are available in the literature for EMG signal segmentation [21]. The segmentation procedure contributes to the accuracy of the classification procedure by extracting accurately the time event (the start and the end times of the event).

The feature extraction step has shown to be a crucial step to the classification procedure as the accuracy of the classification procedure depends on the feature extraction method (Table 1). The implementation of the three different feature extraction methods has proven that time–frequency information was required to correctly classify the vowels. Temporal and spectral features had the same order of accuracy, while wavelet packets based method outperformed the two methods. Numerous numbers of classifiers exist in the literature and it was a challenge to decide which method suited this work. WEKA software provided a flexible platform to test many algorithms as shown in Table 1. Random forest classifier along with wavelet packets features produced the highest classification accuracy of 77%. The confusion matrix shown in Table 2 shows the accuracy for classification the three vowels ‘ا’, ‘ي’, and ‘و’ was 77, 72, and 82%, respectively. The table indicates that the major misclassification was between the letters ‘ا’ and ‘ي’ which contributed largely to the overall error.

A voiceless speech recognition system based on wavelet packets and random forests classifier is proposed for the recognition of Arabic vowels which are the most difficult to pronounce in Arabic language. The proposed system can be used in other possible applications such as the recognition of facial expression of deaf people as it is their major way of communication along with the movement of their hands. Possible improvement can be introduced to the proposed system by improving the segmentation procedure, exploring more feature extraction techniques, and enhancing the classification procedure.

References

1. Al-Jarrah O, Halawani A (2001) Recognition of gestures in Arabic sign language using neuro-fuzzy systems. *Artif Intell* 133:117–138
2. Arjunan S, Weghorn H, Kumar D, Yau W (2006) Vowel recognition in English and German language using facial movement

- (SEMG) for speech control based HCI. In: *Conferences in research and practice in information technology (CRPIT)*, vol 56, pp 13–18
3. Asadpour V, Towhidkhah F, Homayounpour M (2006) Performance enhancement for audio-visual speaker identification using dynamic facial muscle model. *Med Biol Eng Comput* 44 (10):919–930
 4. Bang-hua Y, Guo-zheng Y, Ting W, Rong-guo Y (2007) Subject-based feature extraction using fuzzy wavelet packet in brain-computer interfaces. *Signal Process* 87:1569–1574
 5. Betts B, Jorgensen C (2006) Small vocabulary communication and control using surface electromyography in an acoustically noisy environment. In: *Proceedings of HICSS, Hawaii*
 6. Betts B, Binsted K, Jorgensen C (2006) Small vocabulary speech recognition using electromyography. *Interact Comput* 18:1242–1259
 7. Caudill M (1989) *Neural networks primer*. Miller Freeman Publications, San Francisco, CA
 8. Daube J (2002) *Clinical neurophysiology*. Oxford University Press, Oxford
 9. Englehart K, Hudgins B, Parker P, Stevenson M (1998) Time-frequency representation for classification of the transient myoelectric signal. In: *Proceedings of the 20th annual international conference on engineering in medicine and biology*, vol 5, pp 2627–2630
 10. Fraiwan L, Kasawneh N, Lweesy K (2009) Automatic sleep stage scoring with wavelet packets based on single EEG recording. In: *Proceedings of the international conference on medical informatics and biomedical engineering (ICMIBE)*, Paris, vol 54, pp 513–516
 11. Hall M, Smith L (1998) Practical feature subset selection for machine learning. In: *Proceedings of the 21st Australian computer science conference*, pp 181–191
 12. Hannaford B, Lehman S (1986) Short time Fourier analysis of the electromyogram: fast movements and constant contraction. *IEEE Trans Biomed Eng* 33:1173–1181
 13. Huang C, Chen C, Chung H (2005) The review of applications and measurements in facial electromyography. *J Med Biol Eng* 25(1):15–20
 14. Itoh Y, Uematsu H, Nogata F, Nemoto T, Inamori A, Koide K, Matsuura H (2007) Finger curvature movement recognition interface technique using SEMG signals. *J Achiev Mater Manuf Eng* 23(2):43–46
 15. Jang J (1993) ANFIS: adaptive-network-based fuzzy inference systems. *IEEE Trans Syst Man Cybern* 23(3):665–685
 16. Jou S, Schultz T, Walliczek M, Kraft F, Waibel A (2006) Towards continuous speech recognition using surface electromyography. In: *Proceedings of ninth international conference on spoken language processing (ISCLP 2006)*, pp 573–576
 17. Khushaba R, Al-Jumaily A (2006) Fuzzy wavelet packet based feature extraction method for multifunction myoelectric control. *Int J Biol Life Sci* 2:3
 18. Kim J, Jang W, Bein Z (1996) A dynamic gesture recognition system for Korean sign language (KSL). *IEEE Trans Syst Man Cyber B* 26(2):254–359
 19. Krzanowski W (1988) *Principles of multivariate analysis*. Oxford University Press, Oxford
 20. Li PC, Chiang YY, Tsai KS, Young ST (2005) Genetic algorithm for the efficient selection of disyllabic word lists used in Mandarin speech discrimination tests. *Med Biol Eng Comput* 43 (5):648–657
 21. Manabe H, Hiraiwa A, Sugimura T (2003) Unvoiced speech recognition using EMG-mime speech recognition. In: *Proceedings of HFCS, Ft. Lauderdale, FL*
 22. Orfanidis J (1996) *Introduction to signal processing*. Prentice-Hall, Englewood Cliffs, NJ
 23. Parsons T (1986) *Voice and speech processing*. McGraw-Hill, New York
 24. Perrin E, Berger-Vachon C, Collet L (1999) Acoustical recognition of laryngeal pathology: a comparison of two strategies based on sets of features. *Med Biol Eng Comput* 37(5):652–658
 25. Reaz M, Hussain M, Mohd-Yasin F (2006) Techniques of EMG signal analysis: detection, processing, classification and applications. *Biol Proced Online* 8(1):11–35
 26. Shanabeh T, Assaleh K, AL-Rousan M (2007) Spatio-temporal feature extraction techniques for isolated gesture recognition in Arabic sign language. *IEEE Trans Syst Man Cybern B* 26 (2):254–359
 27. Silverman J (2010) EMG segmentation for wireless access. <http://stargate.uwaterloo.ca/~jzelek/teaching/syde361/ficcdat2007/Full%20Papers/CMBEC30/M0194.pdf>. Accessed 24 June 2010
 28. Staude G, Flachenecker C, Daumer M, Wolf W (2001) Onset detection in surface electromyographic signals: a systematic comparison of methods. *EURASIP J Appl Signal Process* 2:67–81
 29. Stylianou A, Luchies C, Insana M (2003) EMG onset detection using the maximum likelihood method. In: *Summer bioengineering conference*, Key Biscayne, FL, pp 1075–1076
 30. Van Boxtel G (2001) Optimal signal bandwidth for the recording of surface EMG activity of facial, jaw, oral, and neck muscles. *Psychophysiology* 38(1):22–34
 31. Vuskovic M, Du S (2005) Spectral moments for feature extraction from temporal signals. In: *International conference on intelligent computing*, pp 1063–1072
 32. Wang C, Gao W, Xuang Z (2001) A real-time large vocabulary continuous recognition system for Chinese sign language. In: *Proceedings of IEEE pacific rim conference on multimedia*, pp 150–157
 33. Witten I, Frank E (2005) *Data mining: practical machine learning tools and techniques*, 2nd edn. Morgan Kaufmann, San Francisco, CA
 34. Xiao H, Qun Y, Waixi L, Jian Q (2008) Feature extraction of surface EMG signal based on wavelet coefficient entropy. *Proceedings of the 2nd international conference on bioinformatics and biomedical engineering (ICBBE)*, pp 1758–1760
 35. Zecca M, Micera S, Carrozza M, Dario P (2002) Control of multifunctional prosthetic hands by processing the electromyographic signal. *Crit Rev Biomed Eng* 30:459–485